

Questions
What are the limitations of non-conventional energy sources?
What do you understand by greenhouse effect and from where it is generated
Comment on the prospects of fossil fuels in India.
With a neat sketch explain the working of a solar water heater.
Evaluate the effectiveness of a Pyrheliometer in measuring solar radiation based on its construction and working, with a diagram.
Apply the concept of solar heating in a solar water heater. Explain the same with suitable diagram.
Analyse the differences between primary and secondary forms of energy in terms of usage, cost, and efficiency.
Apply the principles of photo voltaic effect to explain the operation of a solar module in a pv panel.
Analyse how direct solar heating is collected in solar concentrating and Fresnel lens collectors comparing their mechanisms and performances.
Analyse how Fresnel lens collector efficiency varies with the change of irradiation?
Investigate the function of absorber material in a solar collector.
Differentiate between concentrating and solar Fresnel lens collectors
With the help of block diagrams explain the operations of grid Interactive SPV systems.
How would you apply the yaw control mechanism to ensure optimal alignment of a Horizontal Axis Wind Turbine (HAWT) with changing wind directions?
Evaluate the importance of cut-in wind speed in wind turbine applications and justify its impact on performance of wind energy
Analyse the function of penstock in hydro power plant
Evaluate the limitation in the development of tidal barrage and give some examples of the same.
Investigate how the head of a hydro power plant is responsible for regulating the efficiency. Also mention some of the high head hydro plant in India.
Discuss the principle of a solar collector. How collector coating can be used to improve the performance of a collector?
Classify different types of solar thermal collectors and show the constructional details of a flat plate collector. What are its main advantages?
Analyse the differences between conventional and non-conventional forms of energy in terms of usage, cost, and efficiency.
Apply the principles of photo voltaic effect to explain the operation of a solar array in a pv panel.
Evaluate the declination angle of the sun on 16 May? (GATE 2015)
Analyse how direct solar heating is collected in cylindrical and Fresnel lens collectors comparing their mechanisms and performances.
Explain the future prospective of the renewable energy.
Explain the I-V characteristics of a solar cell and define fill factor.
What is the importance of MPPT in an SPV system?
What do you understand by indirect solar energy
Discuss various types of drive schemes used in wind turbines.
Explain the yaw control method .
What do you understand by offshore wind firm.
Evaluate the power available from a hydro scheme? Given: Water flow rate is $4 \text{ m}^3/\text{s}$, head is 30 m, efficiency is 85%, density of fresh water is 1000 kg/m^3 .
Evaluate the importance of cut-in wind speed in wind turbine applications and justify its impact on performance and energy generation.

Analyse the function of draft tube in hydro power plant
Explain the stall control method in HAWT to control the wind power generation.
Analyze the power versus wind speed characteristics of a wind turbine with the help of a diagram, explaining key regions and trends.
Analyse the comparative features of Horizontal axis and vertical axis wind turbine in the context of wind energy.
Investigate the future prospective of the Wind energy in Indian energy scenario in next 10 years.
Analyse the limitations of non-renewable energy sources and explain how they affect the efficiency?
Investigate the causes and impacts of the global warming on the Earth's climate system.
Analyse the future prospective of the green energy in India's upcoming energy demand.
Differentiate between Flat plate collector and solar concentrating collectors.
Classify the hydro power plant depending on their head.
What do you understand by pumped storage system in hydro power plant, explain.
Explain the future prospective of the renewable energy.
Explain the mechanism of photoconduction in a PV cell.
What do you understand by energy payback period in solar PV?
What do you understand by direct solar energy, explain with one example.
Comment on the relative features of HAWT and VAWT.
Explain the pitch control method in HAWT.
What do you understand by onshore wind farm? Mention two limitation of it.
With the help of a diagram, discuss the power versus wind speed characteristics of a wind turbine.
Discuss gear schemes used in wind turbines
What are the main advantages in use of biogas? What are their main components?
Evaluate the importance of the solar constant in solar energy studies and justify its standard value.
Evaluate the impact of SF6 gases for global warming.
Explain the concept of direct solar heating, with one example?
Apply the concept of Photoelectric effect in the generation of electricity in a solar cell? (GATE 2013)
Evaluate the effectiveness of a sunshine recorder in measuring solar radiation based on its construction and working, with a diagram.
Apply the concept of solar heating in a solar dryer. Explain the same with suitable diagram.
Compare advantages and disadvantages of concentrating type collector over flat plate collector
Describe the classification of solar cells based on the type of active material used.
Explain the construction and working of the pyranometer with a diagram.
Analyse the limitations of conventional energy sources and explain how they affect the efficiency?
Investigate the causes and impacts of the greenhouse effect on the Earth's climate system.
Analyse the future prospective of the renewable energy in India's upcoming energy demand.
Evaluate the importance of the solar constant in solar energy studies and justify its standard value.
Evaluate the impact of water vapour for global warming.
Explain the concept of direct solar energy, with one example?

Apply the concept of Photoelectric effect in the generation of electricity in a solar cell?						
Discuss stall control used in wind turbines.						
What do you understand by tidal barrage and tidal basin.						
What is approximate value of concentration ratio obtained from a CPC collector?						
Evaluate the significance of the main hurdles in the development of tidal energy.						
Investigate how the head of a hydro power plant is responsible for regulating the efficiency. Also mention different types of hydro plant based on their heads.						
With the help of block diagrams explain the operations of stand-alone SPV systems.						
What is the main advantage of using a glass cover in a box type cooker?						
Draw a schematic diagram of solar pond based electric power plant with cooling tower and explain its working						
How can solar thermal energy be converted into electrical energy?						
Give a neat diagram of a central tower receiver power plant and explain its operation.						
What do you understand by greenhouse effect and what are its consequences						
Write the advantages and disadvantages of concentrating collectors over flat plate type of solar collector.						
With the help of a schematic diagram, explain the working of solar water heating.						
What is the basic difference between an active and passive solar heating system?						
Derive the expression for power output of a hydroelectric plant.						
Explain the pitch control method in HAWT to control the wind power generation.						
Analyze the power versus wind speed characteristics of a wind turbine with the help of a diagram, explaining key regions and trends.						
Analyse the relative features of HAWT on VAWT in the context of wind energy.						
Investigate the future prospective of the Tidal energy in Indian energy scenario in next 10 years.						
Differentiate between Flat plate collector and solar concentrating collectors						
Write the advantages and disadvantages of concentrating collectors over flat plate type of solar collector.						
With the help of a schematic diagram, explain the working of solar water heating.						
What is the basic difference between an active and passive solar heating system?						
With the help of block diagrams explain the operations of solar pond.						
What are the different types of solar energy based on its application?						
Define solar constant and surface azimuthal angle?						
With neat sketch explain the working of solar water heating system.						
Explain the construction and working of the Pyrheliometer with a diagram						
Calculate the declination angle of beam radiation on a photovoltaic array located in Thane (25° 10' N, 78° 61' E) Given the following data:						
Date:October 25,2007 Time:10.30h(LAT) Slope of PV array:18° Surface azimuthal angle:27° $\cos \theta = \sin \phi (\sin \delta \times \cos \beta + \cos \delta \times \cos \gamma \times \cos \omega \times \sin \beta) + \cos \phi (\cos \delta \times \cos \omega \times \cos \beta - \sin \delta \times \cos \gamma \times \sin \beta) + \cos \delta \times \sin \gamma \times \sin \omega \times \sin \beta$						
The cost of major components of solar thermal power plant is specified in the following table along with the life spans.						
<table><tr><td>Item</td><td>Cost in Rs/KW</td><td>Life Period</td></tr><tr><td></td><td></td><td></td></tr></table>	Item	Cost in Rs/KW	Life Period			
Item	Cost in Rs/KW	Life Period				

Heat energy collectors	22000	20
Boiler + steam turbine	11900	10
Electric Generator	6500	10
Accessories and tools	500	5

If the rate of interest is 15% and the maintenance cost is 2% of the capital cost per year. Calculate the capital cost per KW. Calculate life cycle cost (LCC) per KW.

Calculate the angle made by beam solar radiation with the normal to a flat plate collector on May 1st at 9:00 hours (Local Apparent Time – LAT). The solar collector is located in New Delhi at latitude $28^{\circ}35'$ North and longitude $77^{\circ}12'$ East. The collector is tilted at an angle of 36° with the horizontal surface and facing due south.

Calculate the declination angle of beam radiation on a tilted plane surface. The surface is tilted at an angle of 45° from the horizontal and oriented 30° west of south. The location of the surface is Mumbai, which has a longitude of $72^{\circ}49'$ East and latitude of $18^{\circ}54'$ North. The calculation is to be performed for 1:30 PM (Local Apparent Time – LAT) on 15th November.

Analyse how paraboloidal dish collector efficiency varies with the change of temperature?

Investigate the function of glazed plate in a solar collector.

Calculate the value of Air Mass if the value of zenith angle is 90 degree?

Calculate the latitude angle of beam radiation on a photovoltaic array located in Thane ($25^{\circ}10' N$, $78^{\circ}61' E$). Given the following data: Date : October 25, 2007 Time : 10:30 h (LAT) Slope of PV array (β) : 18° Surface azimuthal angle (γ) : 27°